Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/24/2007
PI Name:	Yu, Yongjia Ph.D.		
Project Title:	Impact of HZE Particles Exposure on Proliferation and Differentiation of Human Neural System Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed back to 6/1/2008 per PI (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chuang, Eric (National Institutes of Health) Morrill, Steve (UTMB) wu, Ping (UTMB)		
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Task Description:	Space radiations such as HZE particles pose a great risk of damaging the central nerve system (CNS) of astronauts. Human neural stem cells (hNSCs) are of particular interest since functional NSCs in adult brain are potential targets of HZE damage, and because exogenous hNSCs may provide potential therapeutic source for replacing lost neurons in radiation-damaged brain. The long-term objective of this project is to study the mechanisms underlying HZE particles-induced adverse impact on the proliferation and differentiation of a primary human fetal neural stem cell line, K048. K048 retains self-renewal and multipotential differentiation capabilities in vitro and in vivo. Moreover, a novel in vitro priming procedure appears to direct K048 cells to adopt neuronal differentiation fate. We will test two hypotheses:(1) HZE particles disrupt proliferation and differentiation of human NSCs by activating apoptotic and genotoxic-response pathways. (2) Inhibitory neurogenic micro-environmental cues produced by HZE in rat brain can be overcome by this novel in vitro priming prior to transplantation into brain. We will pursue three specific aims: (1) Determine the impact of 56Fe and gamma-ray irradiation on proliferation and differentiation of K048 cells in vitro. (2) Determine gene expression patterns and apoptotic responses of proliferating and differentiating K048 cells following 56Fe and gamma-ray irradiation. (3) Determine whether in vitro primed K048 cells can differentiate into neurons when grafted into neurogenic and non-neurogenic regions of rat brains damaged by 56Fe and gamma-rays. The results of this project will contribute to a better understanding of cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles. To determine the impact of HZE on proliferation and differentiation of K048 neural stem cells in vitro. To determine the fate of primed K048 cells grafted into 56Fe-irradiated rat brains. In this project, we propose to examine the impact of HZE and low LET radiations on the proliferation and differentiation of K048 human neural stem cell line. The K048 cell model is a unique experimental system not only because of its human origin, but also because this line is not transformed and retains the characteristics of normal neural stem cells. Using this system we will address directly to one of the high priority topics listed in NRA 03-OBPR-07 Research Announcement, namely “What are the cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles?”. The proposed research is also within the scope of “Critical Path Roadmap”, regarding risk number (39) “Degenerative tissue diseases including damage to the central nervous system, cataracts, circulatory diseases, or other late non-cancer risks”, and risk number (41) “Early or acute effects from radiation exposure”. Specifically, our proposed studies address Critical Question (10.05) in radiation, “Are there unique biological effects associated with HZEs?”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this reporting period, we have conducted experiments to determine the effects of 56Fe-irradiation on the differentiation of K048 human fetal neural stem cells. K048 cells were exposed to a single dose (ranging from 0.1 Gy to 2 Gy) of 1000 MeV/n iron ions at the NASA Space Radiation Laboratory at Brookhaven National Laboratory. At different time points post-irradiation, the stem cells were induced to differentiate for 10 days, followed by immunocytochemial quantification of neurons and astrocytes in the population. We found that: (1)56Fe -irradiation inhibited the neuron production in K048 differentiation in vitro in a dose-dependent manner. Approximately 33% of differentiated population from unirradiated K048 cells stained TuJ1 positive, a single dose of 0.3 Gy irradiation reduced this figure to about 20%. Following a singe 2 Gy irradiation, only about 4.5% of surviving stem cells differentiated into neurons. (2) Inhibition of neuronal differentiation of K048 cells by 56Fe-irradiation is persistent. For instance, after receiving 2 Gy of 56Fe-irradiation followed by a 32-day (4 cell passages) proliferation, these K048 cells can only generate neurons at about 70% capability of their unirradiated counterparts. (3) 56Fe-irradiated K048 stem cells generate more astrocytes during differentiation in a dose-dependent manner. Approximately 40% of differentiated population from K048 cells receiving 2 Gy 56Fe-irradiation stained GFAP positive, in comparison to about 20% from unirradiated K048. (4) Effect of 56Fe irradiation on astroglial differentiation diminishes gradually over time. After four cell passages, the progeny of K048 cells that survived 56Fe-irradiation displayed similar capability of generating astrocytes as untreated control cells. In addition, we had started our HZE animal studies. Male rats were irradiated cranially with either a 1.0 Gy or a 3 Gy single dose of 1000MeV/n 56Fe ion radiation at the NSRL at BNL. Ten weeks later, human neural stem cells were implanted into hippocampus of irradiated and control rats. Four weeks later, rats were sacrificed and their brain collected. Currently, we are conducting immonohistochemical analyses on these samples to determine the fate of these grafted stem cells.
Bibliography Type:	Description: (Last Updated: 03/19/2010)
Abstracts for Journals and Proceedings	Yu Y , Gao Y, Wu P. "Impact of gamma-ray and 56Fe ion irradiation on human neural stem cell differentiation." Presented at the 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Investigators' Workshops - Moscow and St. Petersburg, Russia, June 5-9, 2006. Meeting proceedings, 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Investigators' Workshops, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Yu Y, Gao Y, Wu P. "Ionizing radiation inhibits neuronal differentiation of human neural stem cells." Oral presentation at the 53rd Annual Meeting of Radiation Research Society, Philadelphia, PA, November 5-8, 2006. (Mini symposium) Meeting proceedings, 53rd Annual Meeting of Radiation Research Society, November 2006. , Nov-2006